

DARIA MELNIKOVA

R&D SYSTEMS ENGINEER
AI / NEUROTECH / BIOMEDICAL ML
CONTACT

EMAIL
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GITHUB
github.com/Godcomplexx

PORTFOLIO
godcomplexx.github.io/portfolio/

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CORE STACK

AI / ML
PyTorch, TensorFlow, Scikit-learn, XGBoost

SIGNALS
EEG, EMG, audio processing, MNE, feature engineering

VISION
OpenCV, MediaPipe, YOLOv8, medical imaging pipelines

BACKEND
FastAPI, Docker, Celery, RabbitMQ, REST APIs

HARDWARE
Orange Pi, GPIO, edge AI deployment, system integration

TOOLS
React, TypeScript, Unity, Blender, Plasticity, Git

EDUCATION

2025 · Master
Samara State Medical University (SamGMU)
PISH, ARTIFICIAL INTELLIGENCE ENGINEERING

2023 · Bachelor
Samara State Transport University
INFORMATICS AND COMPUTER ENGINEERING

2023 · Additional Education
Samara State Transport University
TRANSLATOR

COURSES

2025 · Neuromatch

2025 · FastAI

SUMMARY

3+ YEARS

R&D systems engineer with 3+ years of hands-on experience in practical AI for neurotechnology and biomedical products. I work across the full path from data acquisition and signal processing to inference, interfaces, and deployable system design, with a product-oriented focus on systems that can move from research into usable clinical and engineering workflows.

EXPERIENCE

SAMGMU · 2023 - PRESENT

R&D Engineer, Research Institute, Samara State Medical University
2023 - PRESENT

- Designed and implemented end-to-end ML pipelines for EEG, audio, and video data.
- Built microservices for real-time signal processing, inference, and biomedical workflows.
- Developed computer vision systems using MediaPipe and YOLOv8.
- Created VR rehabilitation scenarios in Unity with Ultraleap hand tracking.
- Implemented an audio apnea detection pipeline with over 80% accuracy.
- Built MRI processing pipelines from DICOM conversion to segmentation services.
- Designed an embedded AI device for concussion screening on Orange Pi.

SELECTED PROJECTS

2025-2026

AI-based Concussion Screening Device
EDGE AI / HARDWARE / PATENT CO-AUTHOR

Hybrid EEG Speller (P300 + LLM)
BCI / LLM-ASSISTED COMMUNICATION / A-B STUDY

Medical Image Segmentation Service
MEDICAL IMAGING / MICROSERVICES / ASYNC PROCESSING

Epileptic Seizure Prediction
EEG / CNN-LSTM / TRANSFER LEARNING

APNEA
AUDIO ML / ADAPTIVE CASCADE / OVER 80% ACCURACY

Hybrid Emotional Response Analysis
VIDEO + EEG FUSION / BATCH ANALYTICS OUTPUTS

PUBLICATIONS

2024

Machine Learning Pipeline for Automated Detection of Sleep Apnea Episodes
IEEE CNN 2024

Deep Learning-Based Analysis of EEG Biomarkers
Science and Innovations in Medicine, 2024

Portfolio projects and additional implementation details are available at
godcomplexx.github.io/portfolio/.